Personal Project

GAMBLING PLATFORM ANALYTICS

An End-to-End SQL, Data Quality and Business Intelligence Case Study

An analysis of betting activity, financial transactions, player engagement, and product usage using synthetic data.



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Tools: Google BigQuery · SQL · Google Sheets · Looker Studio

Executive Summary

This project explores how data analytics can be used to understand betting activity, financial transaction performance, player engagement, and product usage on a simulated online gambling platform.

The analysis followed an end-to-end workflow using SQL in Google BigQuery for data preparation, data quality checks, and exploratory analysis. Google Sheets was used to prepare summary datasets, and Looker Studio was used to build an interactive dashboard consisting of three pages: Executive Overview, Player Insights, and Product & Operations.

The project covered duplicate removal, missing-value assessment, category standardization, relationship validation, trend analysis, player recency, and session engagement. One notable finding was an unusual concentration of 11,333 betting records at the exact timestamp of 31 December 2025, 23:59:59 UTC. This pattern was documented as a data-quality anomaly requiring further investigation.

The resulting dashboards bring together operational metrics and exploratory findings in a format designed to support business monitoring and further investigation. Financial metrics such as gross gaming revenue were treated as provisional because the underlying payout definitions and business rules had not been independently verified.

The project demonstrates a practical approach to transforming raw data into structured analysis, identifying data limitations, and communicating findings through business intelligence dashboards.

1. Introduction

1.1 Project Background

Online gambling platforms generate data across multiple operational areas, including betting transactions, deposits and withdrawals, player accounts, promotional campaigns, and gaming sessions. Analyzing these data sources can help identify activity patterns, monitor operational performance, and highlight areas that require further investigation.

This project uses a synthetic dataset representing an online gambling platform. It applies SQL-based data preparation and exploratory analysis to transform the raw data into structured summaries and interactive dashboards.

The project is designed as a portfolio case study to demonstrate practical data analytics skills. The findings are illustrative and should not be interpreted as evidence about a real gambling business.

1.2 Business Objectives

The main objectives were to:

1. Assess data quality by identifying duplicate records, missing values, inconsistent categories, and potential anomalies.
2. Analyze betting activity over time, including bet volumes, stakes, payouts, and provisional gross gaming revenue.
3. Examine deposit and withdrawal performance, including approval and decline rates.
4. Explore player activity through acquisition channels, account status, and last-bet recency.
5. Compare session engagement across devices and game categories.
6. Develop interactive dashboards that present key metrics and findings clearly.
7. Identify operational questions and recommendations supported by the available data.

2.1 Data Sources and Scope

The project used seven related synthetic datasets containing player, betting, event, transaction, session, bonus, and campaign information. The raw datasets contained 222,640 records in total.

The data included both categorical and numerical fields, as well as timestamps that enabled time-based analysis. The datasets were stored and queried in Google BigQuery.

Because the data are synthetic, the patterns identified in this project are used to demonstrate analytical techniques rather than to make claims about actual customers or business performance.

Dataset	**Raw Records**	**Main Purpose**
Players	5,000	Player profiles, acquisition channels and account status
Bets	100,050	Betting activity, stakes, odds, results and payouts
Events	2,500	Sporting event information and status
Transactions	60,060	Deposits, withdrawals, refunds and bonus credits
Sessions	35,000	Session duration, devices, game categories and bets placed
Bonuses	20,000	Bonus issuance and redemption
Campaigns	30	Campaign type, dates and budget
Total	222,640	Seven related datasets

2.2 Tools Used

- **Google BigQuery and SQL:** Data exploration, duplicate removal, validation, aggregation, and business analysis.
- **Google Sheets:** Storage and organization of exported summary datasets for dashboard development.
- **Looker Studio:** Interactive dashboards, charts, scorecards, and visual reporting.

The analysis focused on SQL and business intelligence rather than predictive modelling.

3. Data Cleaning and Quality Assurance

Data-quality issue	Finding	Action taken
Exact duplicate bets	50 excess rows	Removed exact duplicate rows into bets_clean
Exact duplicate transactions	60 excess rows	Removed exact duplicate rows into transactions_clean
Inconsistent transaction statuses	Mixed capitalization, such as APPROVED and Approved	Standardized statuses in transactions_standardized
Missing betting stakes	35 records	Retained missing values and excluded them from stake-dependent calculations
Missing payment methods	80 records	Preserved NULLs; use Unknown only as a display label when needed
Primary key quality	No NULL or duplicate primary IDs in the checked tables	Validation checks passed
Relationships between tables	No unmatched foreign keys in the checked relationships	Referential integrity checks passed
Numeric and date checks	No issues found in the specified checks	Kept values unchanged
Unusual timestamp concentration	11,333 bets at one timestamp	Retained records and flagged the anomaly for investigation

3.1 Duplicate Removal

Exact duplicate records were identified in the betting and transaction datasets. The original betting dataset contained 100,050 rows, while the cleaned version contained 100,000 rows. The transaction dataset decreased from 60,060 to 60,000 rows.

Query results

Job information | Results | Visualisation

Row	table_name	duplicate_ids
1	bets	50
2	bonuses	0
3	campaigns	0
4	events	0
5	players	0
6	sessions	0
7	transactions	60

Query results

Job information | Results | Visualisation

Row	table_name	row_count
1	bets_raw	100050
2	bets_clean	100000
3	transactions_raw	60060
4	transactions_clean	60000

The duplicate removal was performed using distinct-row selection. The raw tables were preserved, allowing the cleaned versions to be compared against the original data.

3.2 Missing Values

Thirty-five records in the betting dataset had missing stake amounts. These values were retained as NULL because the available fields did not provide a reliable basis for reconstructing the missing stakes.

Query results Chat View all jobs

Job information | Results | Visualisation | JSON | Execution details

Row	table_name	column_name	missing_count
1	transactions_clean	payment_method	80
2	bets_clean	stake_amount	35
3	bets_clean	bet_id	0
4	bets_clean	bet_time	0
5	bets_clean	bet_type	0
6	bets_clean	bonus_used	0

Eighty transaction records had missing payment methods. These values were also retained rather than assigned an assumed payment method. Where appropriate, reports can display these missing values under an 'Unknown' label.

3.3 Category Standardization

Transaction status values were inconsistent in capitalization. The standardized table consolidated equivalent values into Approved, Declined, and Pending categories. This improved consistency in subsequent transaction reporting.

3.4 Validation

Checks were conducted for primary key uniqueness, missing primary IDs, relationships between related datasets, negative numeric values, invalid decimal odds, negative session durations, and campaign date inconsistencies.

No issues were found in the specific numeric and relationship checks performed, and the checked primary keys were unique and non-null.

3.5 Anomaly Detection

A major concentration of betting records was identified on 31 December 2025. Of the 11,596 recorded bets that day, 11,333 shared the exact timestamp 23:59:59 UTC.

Query results

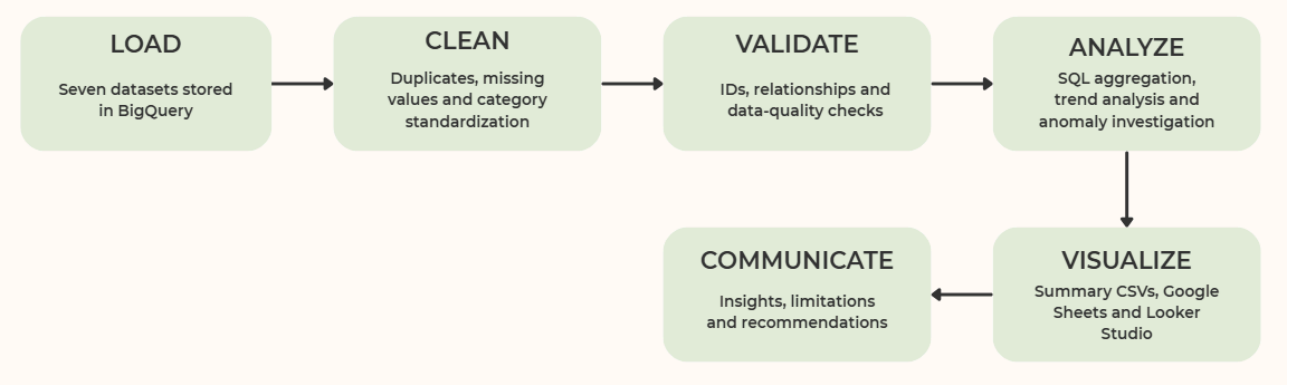
Chat | View all jobs | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	month	days_with_bets	avg_daily_bets	min_daily_bets	max_daily_bets	monthly_bets
1	2025-11	30	234.7	209	264	7040
2	2025-12	31	618.9	226	11596	19185

The records were retained because the cause had not been confirmed. The concentration may indicate a timestamp or data-generation issue, but further investigation would be required to establish its origin. The anomaly was documented as a limitation when interpreting time-based betting trends.

4. Analytical Methodology

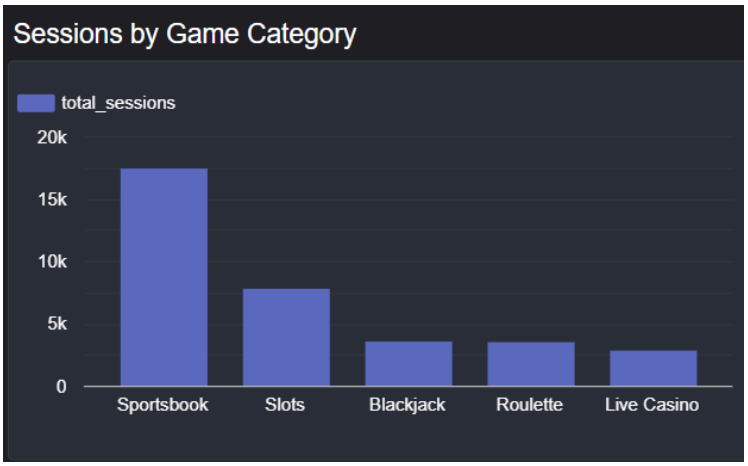
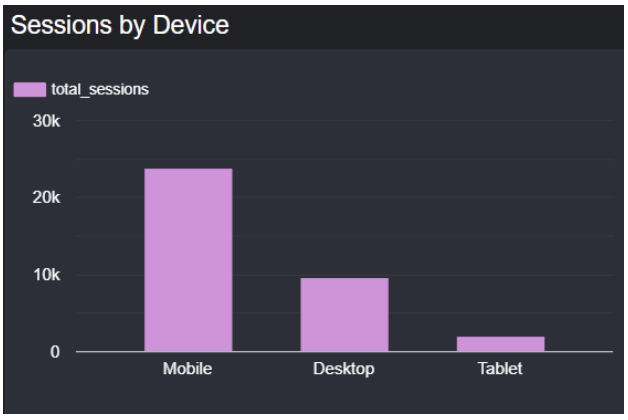


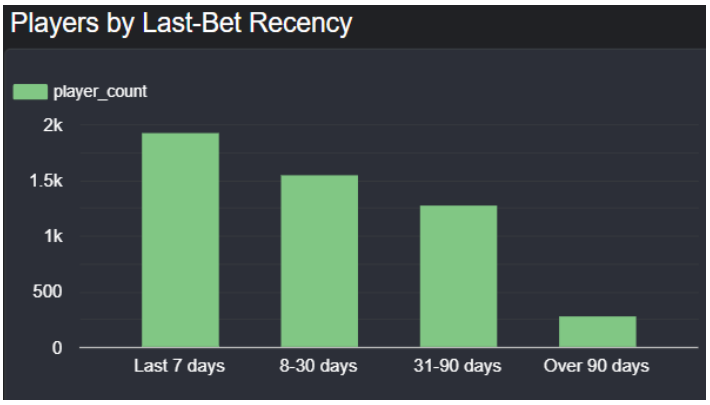
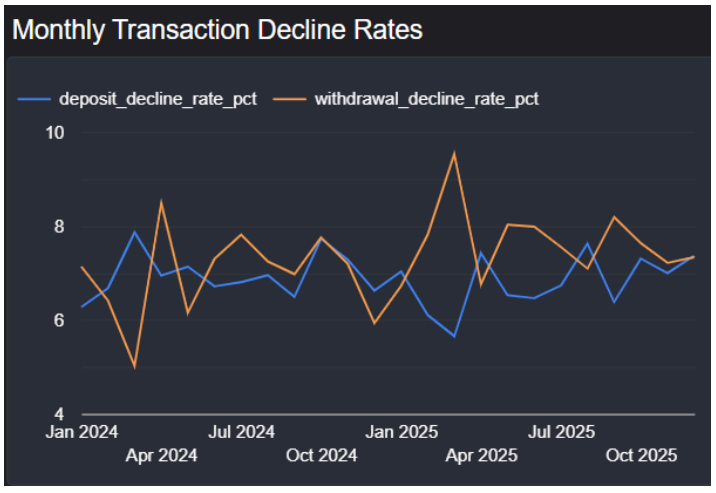
The analysis followed a structured workflow.

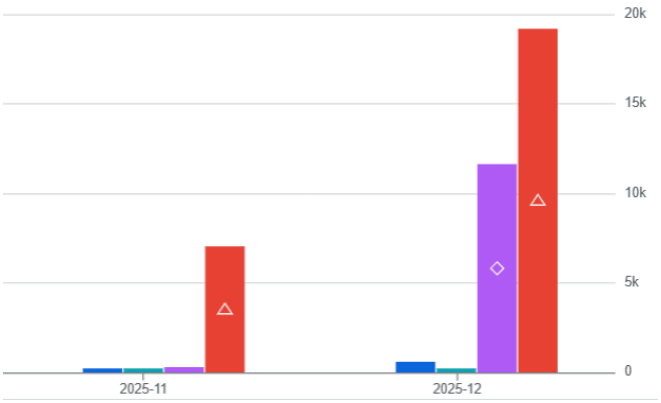
1. **Data preparation:** Loaded and organized the seven datasets in BigQuery.
2. **Data cleaning:** Removed exact duplicate records from the betting and transaction datasets and standardized inconsistent transaction statuses.
3. **Data validation:** Checked primary keys, table relationships, missing values, and selected numerical and date constraints.
4. **Exploratory data analysis:** Used SQL aggregation and grouping to examine betting trends, transaction performance, player recency, and session engagement.
5. **Anomaly investigation:** Investigated an unusually high betting volume on 31 December 2025, drilling down from daily to hourly, minute-level, and second-level counts.
6. **Dashboard development:** Exported summary query results to CSV, organized them in Google Sheets, and connected the worksheets to Looker Studio.

The analysis was descriptive and exploratory. It identified patterns in the available data but did not establish causal relationships or validate the financial definitions used by a real platform.

5. Key Findings & Business Insights

49.72% Sessions	**Finding 1 — Sportsbook leads session volume** Sportsbook recorded 17,403 sessions, accounting for approximately 49.72% of all 35,000 sessions. Slots ranked second, with 7,766 sessions (22.19%). Business implication: Sportsbook deserves priority when investigating session volume, usability, and operational performance. Session volume alone does not establish profitability. 
67.64% Sessions	**Finding 2 — Mobile dominates device usage** Mobile accounted for 23,673 sessions, compared with 9,470 on desktop and 1,857 on tablet. Business implication: Mobile experience and reliability should be considered a priority. Average session duration remained close to 40 minutes across all three devices. 

69.18% Players	**Finding 3 — Most players bet within 30 days of the dataset reference date** 3,459 of the 5,000 players had their last recorded bet within 30 days of the dataset's latest betting timestamp. Another 1,541 players fell into the over-30-day groups. Business implication: Recency can help prioritize further engagement analysis, but the measure is relative to the dataset date and does not establish current activity or churn. 
+0.70 pp Year-over-year	**Finding 4 — Withdrawal decline rate increased in 2025** The annual withdrawal decline rate rose from approximately 6.95% in 2024 to 7.65% in 2025. The deposit decline rate decreased slightly, from 6.96% to 6.81%. Business implication: Investigate withdrawal decline reasons, payment methods, and processing outcomes. The current dataset does not identify the cause of the increase. 

11,333 Bets	**Finding 5 — A major timestamp anomaly affected December 2025** A total of 11,333 betting records shared the same second-level timestamp on 31 December 2025. This accounted for approximately 97.7% of that day's 11,596 bets. Business implication: Verify the source timestamps and data-generation process before using the affected day for operational comparisons or time-based conclusions. 

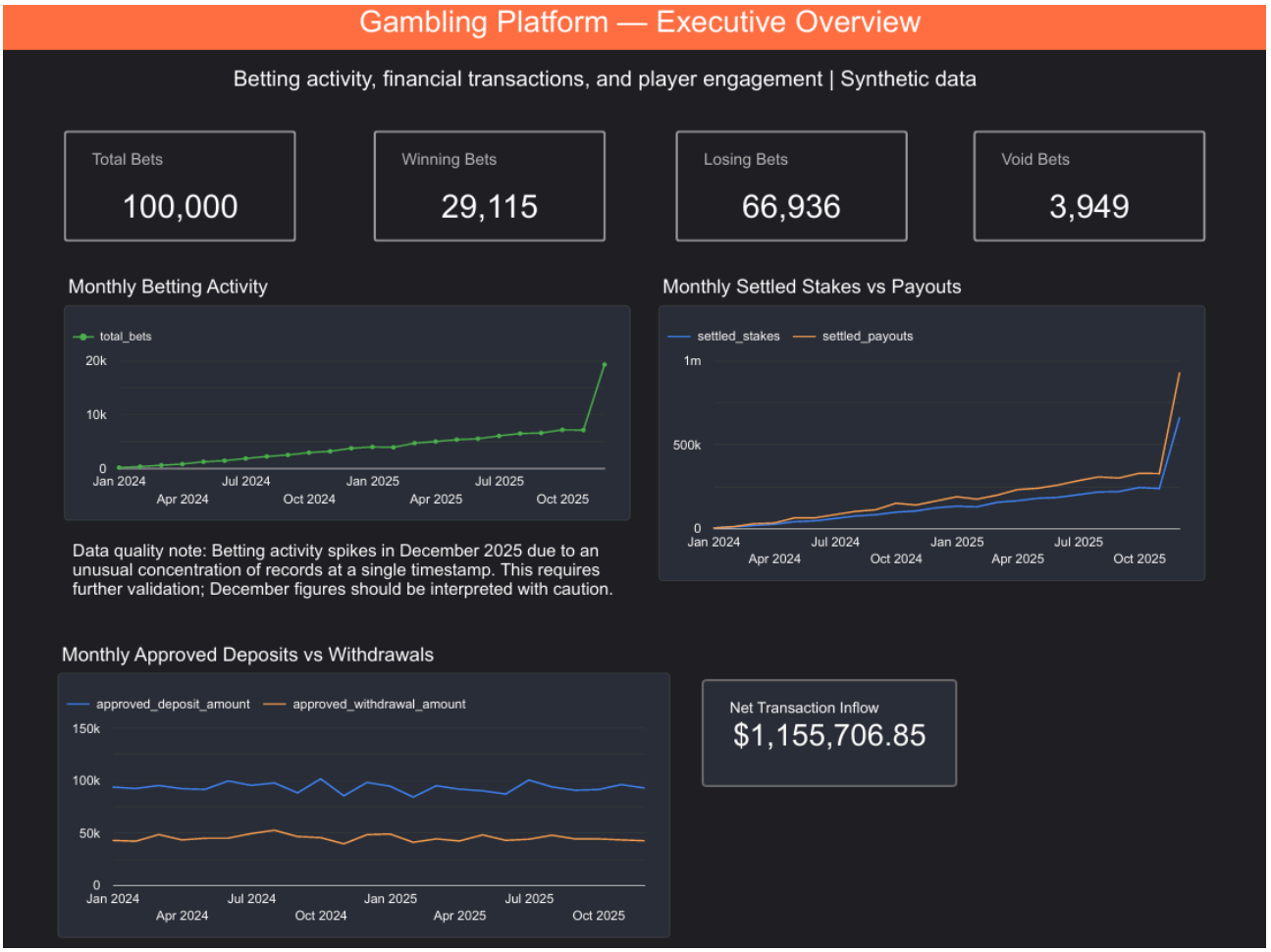
6. Dashboard Development

The analysis was presented through a three-page Looker Studio dashboard. Each page focuses on a different aspect of platform operations.

6.1 Executive Overview

The Executive Overview brings together headline betting metrics, monthly betting activity, settled stakes and payouts, and approved deposit and withdrawal amounts. It is intended to provide a high-level view of operational activity over time.

A separate data-quality note highlights the unusual concentration of betting records on 31 December 2025. Financial metrics are presented cautiously because the dataset's payout definitions and financial business rules have not been independently verified.



6.2 Player Insights

The Player Insights page presents player recency and acquisition-channel information. These views help describe historical activity patterns and the distribution of registered players across acquisition sources.

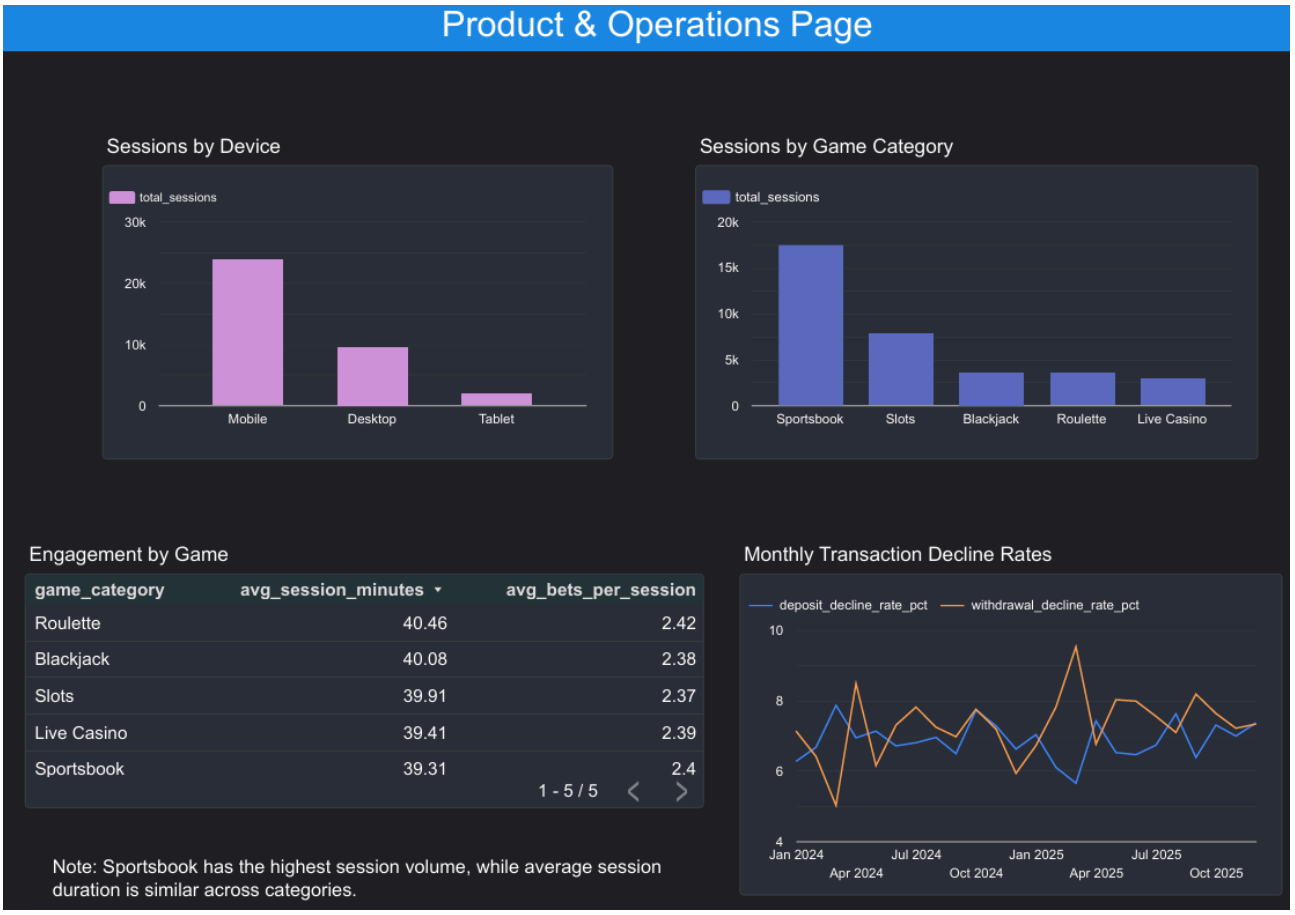
The recency categories use the latest betting timestamp in the dataset as their reference point. They should not be interpreted as live engagement status or confirmed churn. In addition, every player in the synthetic dataset had at least one recorded bet, limiting the usefulness of historical betting conversion as a comparison between acquisition channels.



6.3 Product and Operations

The Product & Operations page compares session volume and engagement across devices and game categories. It also displays monthly transaction decline rates.

Sportsbook generated the largest number of recorded sessions, and mobile accounted for the majority of sessions across devices. Transaction decline trends provide a starting point for further investigation, but additional data would be required to determine why transactions were declined.



6.4 Dashboard Design Principles

The dashboards were organized around business questions, with clear chart titles and summary metrics. The objective was to make the results easy to interpret while preserving important data-quality caveats and avoiding unsupported financial conclusions.

7. Recommendations

Based on the exploratory findings, the following actions are recommended for further investigation:

1. **Investigate the timestamp anomaly.** Review the source data and timestamp-generation process for the 11,333 records sharing the same timestamp on 31 December 2025. Retain the records until the issue is understood.
2. **Review withdrawal declines.** Compare the higher withdrawal decline rate in 2025 with payment-method performance, transaction outcomes, and decline reasons if those fields become available.
3. **Prioritize mobile experience monitoring.** Given mobile's share of session volume, examine mobile performance, errors, usability, and session reliability.
4. **Investigate player recency.** Combine last-bet recency with reliable account-status dates and additional activity data before defining inactivity or churn.
5. **Validate financial metrics.** Confirm how payouts, void bets, bonuses, refunds, and transaction amounts should be treated before using the measures for financial reporting.

6. **Improve monitoring and data documentation.** Maintain a record of cleaning decisions, validation results, metric definitions, and data-quality issues.

These recommendations are intended as next steps for investigation, rather than claims that a particular business problem or cause has been confirmed.

8. Limitations

- **Synthetic data:** The datasets are simulated and may not reflect real player behavior or business operations.
- **Financial definitions:** Gross gaming revenue was calculated provisionally using stakes minus payouts for settled bets with known stakes. The definition has not been validated against an authoritative financial specification.
- **Missing values:** Missing stake amounts and payment methods were retained rather than inferred.
- **Timestamp anomaly:** The unusual concentration of bets at one timestamp may distort daily and monthly activity comparisons.
- **Limited explanatory fields:** The data do not establish why transactions were declined or why player activity changed.
- **Historical recency:** Player recency was measured relative to the latest recorded bet timestamp, not the current date.
- **Descriptive analysis:** The project identifies patterns but does not establish causation or demonstrate the financial impact of any recommendation.

9. Conclusion

This project demonstrated an end-to-end data analytics workflow using synthetic data representing an online gambling platform. SQL in BigQuery was used to clean, validate, and analyze related datasets, while Google Sheets and Looker Studio were used to organize summary results and develop interactive dashboards.

The analysis identified differences in session volume across devices and game categories, changes in transaction decline rates, historical player recency patterns, and a significant timestamp anomaly in the betting data. These findings illustrate how data analytics can help organize operational information, highlight patterns, and identify areas that require further investigation.

The project also demonstrated the importance of data quality, transparent metric definitions, and careful interpretation. In particular, missing values and anomalous records were not silently corrected, and financial measures were treated cautiously where business definitions remained unverified.

Overall, the project strengthened practical skills in SQL, data cleaning, exploratory analysis, data validation, and dashboard development. Future improvements could include automated data refreshes, more detailed transaction diagnostics, validated financial metrics, and additional monitoring of unusual data patterns.